

Internet of Things

Mobility in IoT

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MSc in Web Intelligence





Overview

- Sink Mobility Protocols
- Adaptive Sink Trajectories
- Sink & Sensor Mobility



Limitations of Static Sink(s)

- Computation and communication (energy) overhead *on the sensors*
- *Low Scalability*
- *Uneven load distribution* among the sensors
- Bypassing obstacles requires a lot of resources
- Dynamic networks have *significant reconfiguration cost*



Mobility: Main Idea

Advances in technology and new applications suggest that sensors may be mobile.

- Wildlife monitoring
- Monitoring in urban environments
- Drones - UxVs

Introducing mobility *shifted the burden from the sensor nodes to the sink*

Main Idea:

- Sink has significant and easily replenished energy reserves.
- The sink can move inside the sensor network area, in close proximity to the sensors.
- By travelling in the whole network area, sink collects all the available data.



Mobility: New Challenges

Sink mobility in WSNs presents *many new challenges*

- Sink must *cover* the whole network
- Incurs *longer delivery delays*
- *Bad scalability* when network area increases
- Routing and localization problems become *more difficult*



Mobility: Advantages

- *Sparse, disconnected and irregular networks* can be better handled
- The mobile sink can *bypass obstacles*
- *Better load distribution*
- *Scales well* with respect to number of sensors
- Reduces communication distance
- *Reduces energy consumption on the sensors* $\Rightarrow$ System lifetime increases
- Reduces adversarial overhearing $\Rightarrow$ *Enhances security*



Early Works on WSNs with Mobility

- ❶ J. Luo and J.-P. Hubaux. *"Joint Mobility and Routing for Lifetime Elongation in Wireless Sensor Networks."* In 24th IEEE INFOCOM, 2005. (optimal movement)
- ❷ D. Jea, A. Somasundara, and M. Srivastava. *"Multiple controlled mobile elements (data mules) for data collection in sensor networks"*. (deterministic movement) In DCOSS 2005.
- ❸ D. Goldenberg, J. Lin, A. Morse, B. Rosen, and Y. Yang: *"Towards mobility as a network control primitive."* In MobiHoc 2004. (optimal placement)



Joint Mobility and Routing for Lifetime Elongation in Wireless Sensor Networks

The main idea is that a mobile entity can help remove “hot spot” areas, reduce energy consumption and better equalize the load among the sensors.

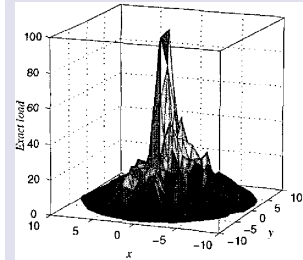
- sensors are uniformly distributed in a circle of radius R
 - ▶ strongly connected network
 - ▶ full coverage
- sensors generate data at a constant rate λ
- shortest path routing assumed

The authors propose an analytical model for calculating the load on each sensor.

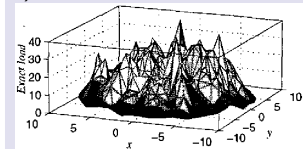


Results:

- the optimal movement trajectory is a circle along the periphery of the network
- optimum routing switches between shortest path and trajectory based routing
- arbitrary motion can reduce maximum load by 3 times
- joint mobility and routing can increase lifetime by 500%



a) Static sink



b) Mobile sink



D. Jea, A. Somasundara, and M. Srivastava

Multiple controlled mobile elements (data mules) for data collection in sensor networks

The main idea is that multiple mobile sinks can be used to fully cover the sensor network.

- Sinks move deterministically
 - ▶ linear parallel trajectories
 - ▶ all nodes are within one hop from *at least one* and *no more than two* sinks
 - ▶ nodes that can communicate with two sinks are called *shareable*
- All data is transmitted over one hop.
- Data throughput is maximized by equalizing the load among the sinks.

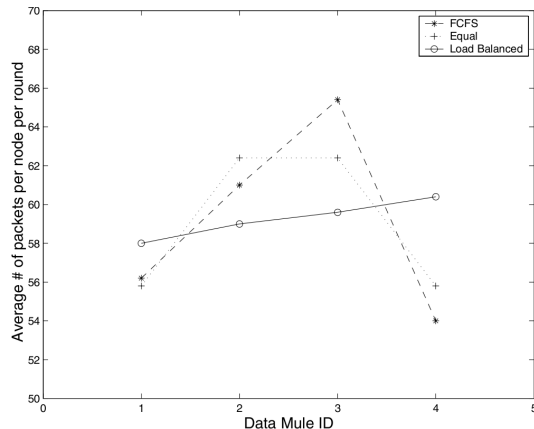
Example

Region A	Data Mule M1
Region B	
Region C	
Region D	Data Mule M2
Region E	
Region F	
Region G	Data Mule M3
Region H	



Three load balancing protocols

- **First Come First Serve:** Shareable nodes associate to the first sink they listen
- **Equal share:** Shareable nodes are equally shared between the two sinks
- **Load balanced:** Shareable nodes are assigned according to the total load of each sink



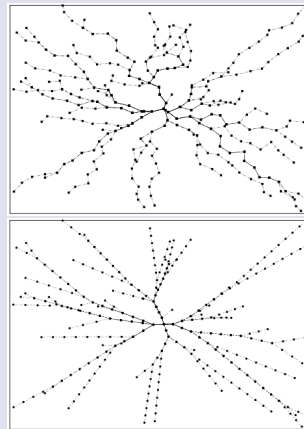


Towards mobility as a network control primitive

- Nodes can move
 - ▶ controlled movement
 - ▶ localization is available
- Optimize routing paths wrt energy
- Examine constrained vs unconstrained movement

The authors prove that the optimal arrangement of nodes is along a straight line in equally spaced positions.

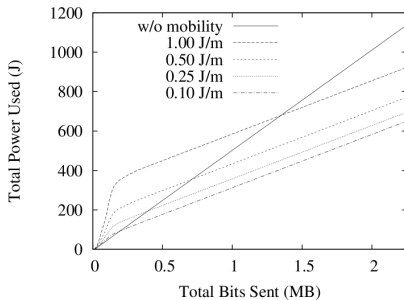
Example



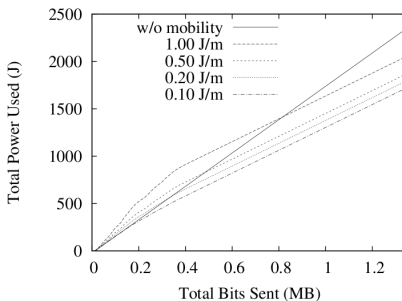


Goldenberg et al

- Initially a lot of energy is spent for optimizing the path
- Optimizing paths saves significant amounts of energy in the long term
- Unconstrained movement spends more energy on movement but saves more energy in the long term



a) unconstrained



b) constrained



Simple Sink Mobility

Athanasios Kinalis, Sotiris E. Nikolettseas, Dimitra Patroumpa, José D. P. Rolim: ***Biased sink mobility with adaptive stop times for low latency data collection in sensor networks***. *Information Fusion* 15: 56-63 (2014)

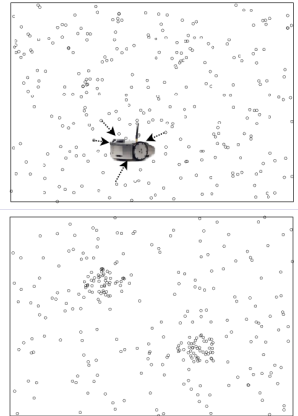
- One of the first approaches to sink mobility, particularly with *randomized techniques*
- Authors assume a *very weak model* i.e. limited or no network knowledge, no coordination of sensors
- Authors present several *combinations of mobility strategies and data collection techniques*, achieving *trade-offs mainly between energy consumption and latency*
- The protocols *significantly reduce energy dissipation* and maximize the success rate; however *latency increases*



The Model

- A number of n ultra-small homogeneous sensor devices are spread in an area $\mathcal{D} \times \mathcal{D}$
 - ▶ random uniform deployment
 - ▶ "clustered" deployment: p_n "pockets" i.e. regions in the network area with high particle density d_p
- Sensor devices do not move

Network Model





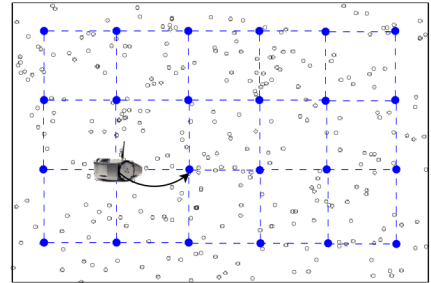
P1: Random Walk with Passive Data Collection

- S performs random walk: $\mathcal{M}_{random}$ selects a random direction and a random distance
- S periodically broadcasts beacons
- Sensors cache the recorded data
- When sensor hears a beacon starts to transmit the cached data
- *Very low communication overhead on sensors*
- *Requires very little network knowledge*



P2: Partial Random Walk with Limited Multihop Collection

- $\mathcal{S}$ partitions the rectangular network area in $j \times j$ square regions
- Each region is represented by a vertex in an overlay graph $G_o(V, E)$ stored *locally* on the $\mathcal{S}$
- $\mathcal{M}_{graph}$: $\mathcal{S}$ moves from $v_i \in V$ to another region v_j adjacent to v_i selected *randomly and uniformly*
- $\mathcal{S}$ periodically broadcasts beacons with a hop counter h_c and a *ttl*
- Beacons form minimum hop propagation trees *ttl* hops deep





P3: Biased Random Walk with Passive Collection

- $\mathcal{S}$ partitions the rectangular network area in $j \times j$ square regions
- $\mathcal{S}$ constructs locally an overlay graph $G_o(V, E)$
- Each region is associated with a *visit counter* c_v and a *density counter* d_v
- Each time $\mathcal{S}$ visits a region, it updates the counters
 - ▶ $c_v = c_v + 1$
 - ▶ $d_v = d_v + \# \text{ unseen before sensors}$
- $\mathcal{M}_{bias}$: moves from the center of one region to the center of another adjacent region
- The next region is selected with “*biased*” probability p_v



P3: Biased Random Walk with Passive Collection (cont.)

Frequency of visits

- $c_{\text{neigh}}(u) = \sum_v c_v$ for all $v : (u, v) \in E$.
- $p(f)_v = \frac{1 - c_v / c_{\text{neigh}}(u)}{\deg_u - 1}$
- *Favours less frequently visited areas*

Local Density

- $d_{\text{neigh}}(u) = \sum_v d_v$ for all $v : (u, v) \in E$
- $p(d)_v = \frac{1 + d_v / d_{\text{neigh}}(u)}{\deg_u + 1}$
- *Dense areas are favoured*

Weighted Combination

$$p_v = \alpha \cdot p(f)_v + \beta \cdot p(d)_v$$

where $\alpha \geq 0$, $\beta \geq 0$ and $\alpha + \beta = 1$.



P3: Biased Random Walk with Passive Collection (cont.)

- Sensors transmit the data when they receive a beacon
- Low communication overhead on the sensors
- Frequency bias increases *fairness*
- Density bias *better copes with irregular node deployment*



P4: Deterministic Walk with Multihop Collection

- $\mathcal{S}$ follows a *static path* of length l
- $\mathcal{M}_{circle}$: moves on a circle of radius $r = \frac{l}{2\pi}$
- $\mathcal{M}_{line}$: moves back and forth on a line segment of length l
- $\mathcal{S}$ transmits beacons that form unlimited minimum hop propagation trees
- Sensors store their hop distance h_d from the sink
 - ▶ if $h_d > 0$ propagate data to the next hop
 - ▶ if $h_d = 0$ propagate data to the sink or cache if sink not present
- *Significant communication overhead* on the sensors



Findings Overview

All protocols achieve *high success rate* and *low energy consumption* at the cost of *increased latency*.

- P1 – Random Walk with Passive Data Collection
 - ▶ Increasing mobility speed improves performance significantly
 - ▶ Unaffected by network topology
- P2 – Partial Random Walk with Limited Multihop Collection
 - ▶ Increasing the size of the data collection area significantly *reduces delay* but also reduces success rate
- P3 – Biased Random Walk with Passive Collection
 - ▶ Frequency biased movement $\simeq$ 100% success rate, low energy consumption and *low delay*
 - ▶ Density biased movement can improve performance in *very dense areas*
- P4 – Deterministic Walk with Multihop Collection
 - ▶ Protocol performance is greatly affected by *l*
 - ▶ *Circular paths are slightly better than linear paths*

Constantinos - Marios Angelopoulos, Sotiris Nikolettseas, Dimitra Patroumpa, and Jose Rolim.
Coverage-Adaptive Random Walks for Fast Sensory Data Collection
ADHOC-NOW 2010 - Edmonton, Canada



Random Walks in WSNs

Random walks can serve as *fully local, very simple strategies* for sink motion, *reduce energy dissipation* a lot but increasing latency.

To achieve satisfactory energy-latency trade-offs the sink walks can be made adaptive, depending on local network parameters such as density and/or history of past visits in each network region.



The Network Model

- Planar area
- Sensors deployment is random uniform over the network area
- All sensors have sensory data to deliver
- No data is generated during the network traversal (focus on data collection)
- Visited nodes are distinguished by lack of data.



The Sink

- During the network initialization, a graph formation phase is executed by the sink
- The network area is partitioned in $j \times j$ equal square regions, called cells.
- A virtual lattice graph $G_o = G(V, E)$ is created which is overlaid over the network area.
- When the sink is located at the center of a cell, it can communicate with every sensor node within the cell area.



Blind Random Walk 1/2

- The most simple and straightforward of all.
- Each move is stochastically independent from all previous ones
- The sink selects its next position with the same probability for each one of the four coordinate directions.



Blind Random Walk 2/2

- Very robust solution
- Probabilistically guarantees that eventually all the cells of the network will be visited
- All data will be collected

However, in some network structures it may become inefficient, mostly with respect to latency.



Random Walks with Memory 1/2

Extends the Blind Random Walk. Uses some (constant) memory of past visits.

- The sink maintains a First-in-First-out (FIFO) list M
- M contains the last K cells visited during the random walk.
- The next hop is chosen uniformly among the neighbours of the cell that are not in the memory list M



Random Walks with Memory 2/2

Trade-off: The use of memory eliminates loops in random walks, but it may also lead to a deadlock.

- If $K = 0$, the random walks become blind and can have loops but no deadlocks.
- For complete memory, the random walks can only have deadlocks and no loops.
- When the size of M is $0 \leq K \leq n - 1$, random walks can have both loops and deadlocks.



Random Walk with Inertia 1/2

- Four directions: North, South, East and West
- Each one is assigned a probability.
- the probability distribution on each step of the walk changes adaptively to the nodes' discovery

The following principle is followed:

Reinforce the direction where newly discovered sensors were found and weaken the directions where already visited sensors have been identified.



Random Walk with Inertia 2/2

p_c : current direction probability at time t is:

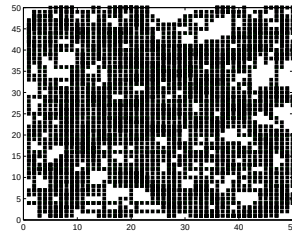
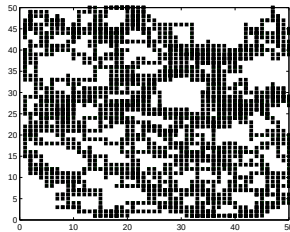
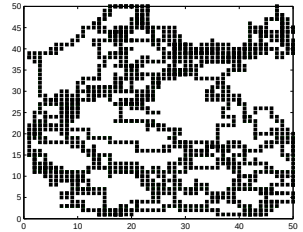
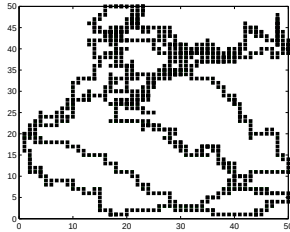
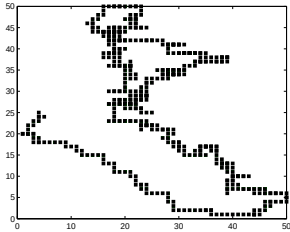
$$p_c^{t+1} = \begin{cases} p_c^t + \delta & , \text{ if new nodes discovered} \\ p_c^t - \delta & , \text{ if no new nodes discovered} \end{cases}$$

while each one of the probabilities towards the rest three directions (p_r : rest direction probabilities) are:

$$p_r^{t+1} = \begin{cases} p_r^t - \frac{\delta}{3} & , \text{ if new nodes discovered} \\ p_r^t + \frac{\delta}{3} & , \text{ if no new nodes discovered} \end{cases}$$



Visualization of the Inertia Walk





Explore-and-Go Random Walk

The sink motion consists of two types of motion:

- moving on a straight line
- arbitrarily changing direction

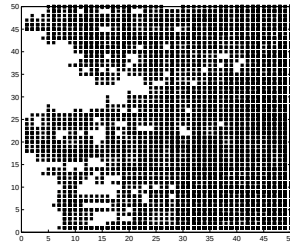
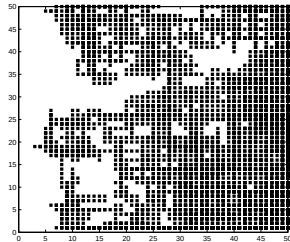
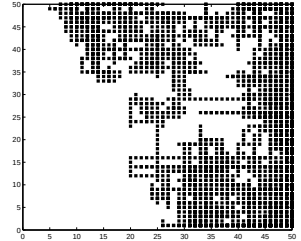
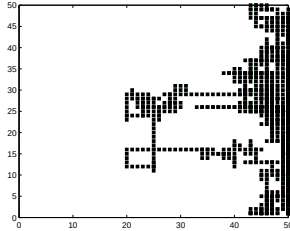
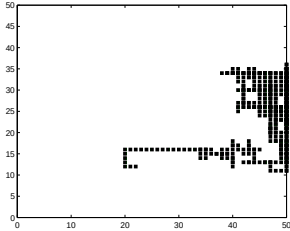
The sink chooses between them via a *bias factor* β .

$$F_{motion} = \begin{cases} \text{move straight, with probability } \beta \\ \text{change direction, with probability } 1 - \beta \end{cases}$$

$$\text{where, } \beta = \begin{cases} 0.1 & , \text{ when new nodes were discovered} \\ 0.9 & , \text{ when no new nodes were discovered} \end{cases}$$



Visualization of Explore-n'-Go





Inertia vs Explore-n'-Go

Similarities

- Require zero knowledge of network area
- Require zero knowledge of sensor distribution.
- Have light-weight requirements in terms of memory and computational power

One main difference:

- Inertia walk performs network coverage by drawing big straight lines.
- Explore-n'-Go performs a more systematic network coverage, by sequentially visiting network subregions .



Curly Random Walk 1/3

Intuition:

- start by visiting a confined subregion
- gradually allow the sink to perform a motion of higher degree of freedom
- eventually, the sink will cover the entire network area.

To achieve this, initially the sink performs frequent narrow left-turns, which gradually get wider.



Curly Random Walk 2/3

This type of motion can be modelled as a series of successive straight S and left- turn L moves.

$SLSSLSSSLSSSSL...$

Obviously, the probability distribution of left turns is the geometric distribution.

- probability mass function $P^i = (1 - p_L)^i p_L$
- p_L : probability of left turn
- i the number of successive straight moves before the next left turn.
- For a fixed i , $p_L^i = \frac{1}{i+1}$.



Curly Walk 3/3

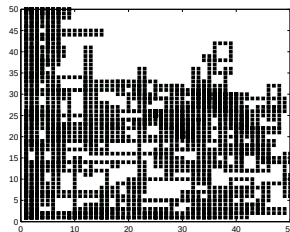
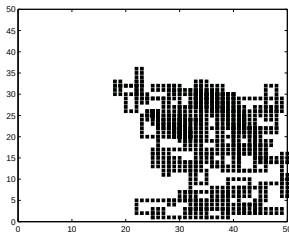
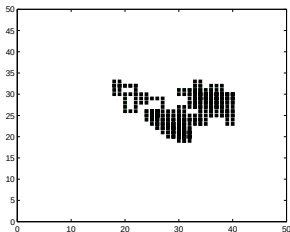
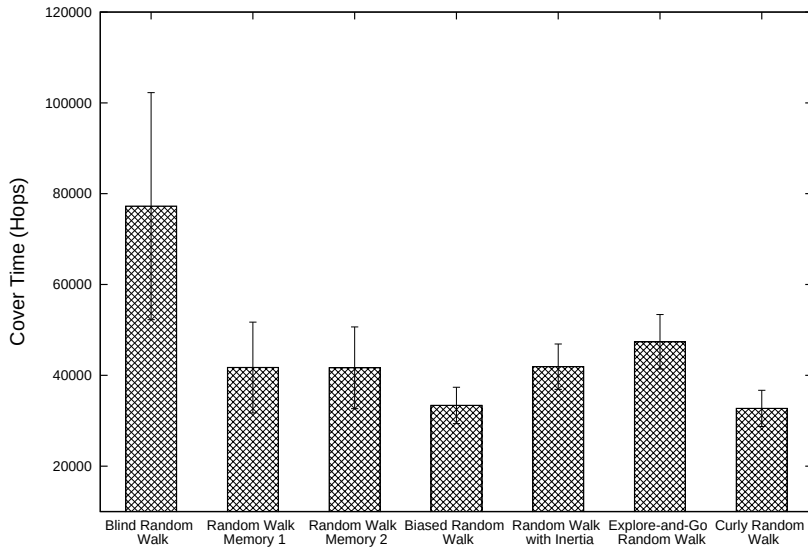


Figure: An example of network traversal following Curly Random Walk in a 50×50 network area. (snapshots after 500, $2 * 10^3$ and $2 * 10^4$ hops.)

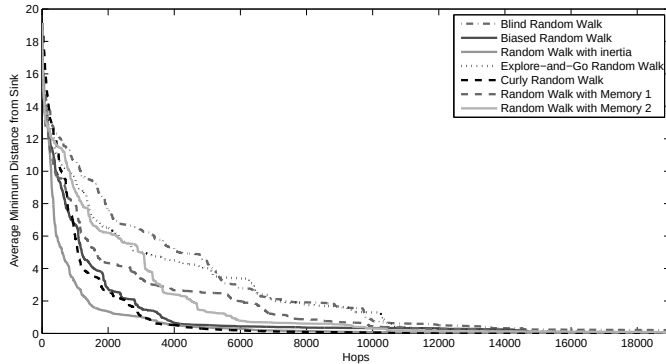


Cover Time





A New Metric: Proximity Variation



The mean value (over all cells) of the smallest distance from the sink for all the cells.

$$PV = \frac{\sum_{i=1}^n \min(dist(i))}{n}$$

Angelopoulos Constantinos Marios, Nikolettseas Sotiris

Aggregated Sensory Data Collection by Mobility-based Topology Ranks

IEEE GLOBECOM 2009

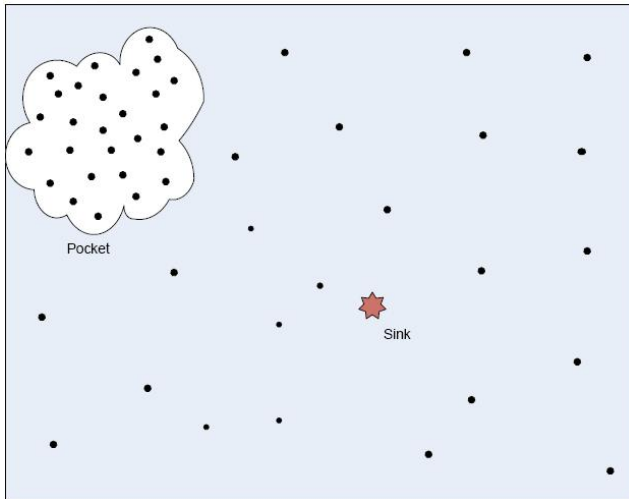


The Problem

In a WSN with *full mobility scheme*, where both sensors and sink move dynamically, how can the sink *efficiently* collect data from sensors?



The Problem





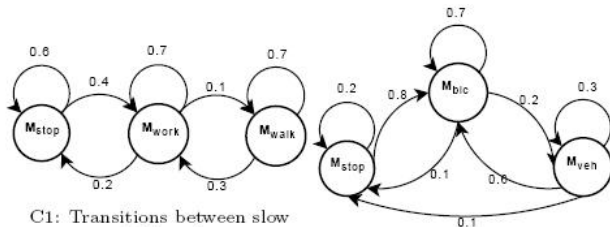
The Sensor Model

Both the sensors and the sink are assumed to be mobile and equipped with (*relative*) localization hardware

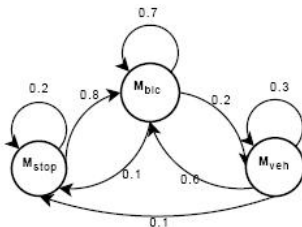
- a $D \times D$ plane network area
- n total number of sensors deployed
- d is the density of sensors in that area (sensors/ m^2)
- Sensor devices are equipped with a set of hardware monitors that can measure environmental conditions of interest.
- also, fixed transmission range R , general purpose storage memory of small constant size



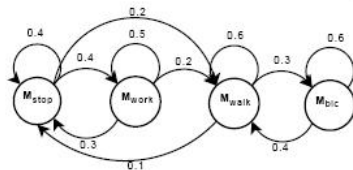
Modelling Dynamic Sensory Mobility



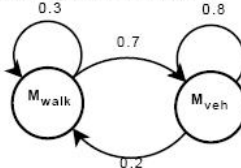
C1: Transitions between slow mobility roles



C3: Transitions for medium mobility with fast bursts



C2: Transitions for medium mobility level



C4: Transitions for fast mobility



Our Approach

- Sensors are moving inside the network area, independently of each other.
- Each sensor periodically measures the number of its neighbours and stores a tuple.
- Each tuple consists of: the number of neighbours, a timestamp, the current position

Based on tuples, the sink is going to exploit general topological information.



Our Approach II

Each tuple is assigned a value via the ranking function

$$R = \frac{d_{local}^2}{\Delta P \Delta T}$$

where:

- d is the number of neighbours.
- ΔP the distance between position where d was measured and current position.
- ΔT the time interval when d was measured and current time.



Our Approach II

Each tuple is assigned a value via the ranking function

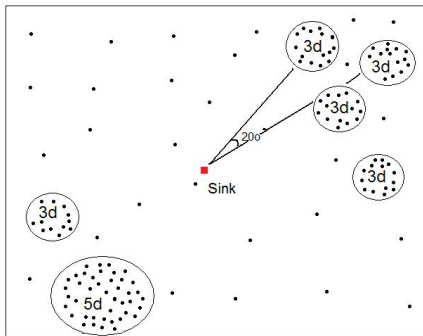
$$R = \frac{d_{local}^2}{\Delta P \Delta T}$$

- Only one tuple is stored each time.
- A new tuple replaces stored one in sensor memory, **iff** it is assigned a higher value.
- When a sensor reaches the radio range of the sink, along with the sensory data, the stored tuple is also sent.
- The sink chooses to move towards the direction corresponding to the highest ranking collected tuple.



The Aggregation Process

The sink initially traverses the network area at random direction. For a short period of time it collects tuples. Tuples A,B corresponding to positions relatively close are aggregated into C, based on an $angle_{thresh}$



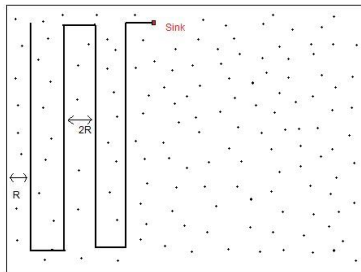
- $C_{d_{local}} = A_{d_{local}} + B_{d_{local}}$
- $C_P = \frac{A_P * A_{d_{local}} + B_P * B_{d_{local}}}{A_{d_{local}} + B_{d_{local}}}$
- $C_T = \frac{A_T * A_{d_{local}} + B_T * B_{d_{local}}}{A_{d_{local}} + B_{d_{local}}}$



Protocols we compare with

Blind Random Walk: In this scenario the sink is simply moving according to a random walk process and serves any sensors that it may reach.

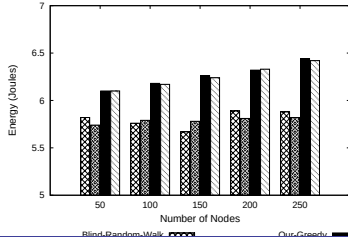
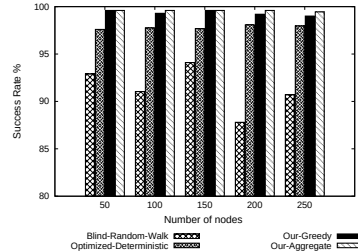
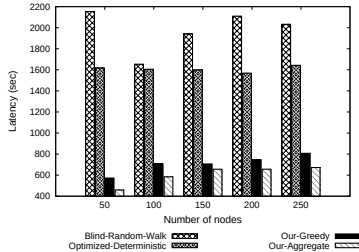
Optimized Deterministic: The sink sweeps the entire area in a way that no overlaps occur.



Our Protocol without Aggregation Process
In order to inspect the impact of the process



Performance Findings



- Latency improves up to 8 times
- Better success rate, from 93% to 98%
- Slightly (10%) more energy dissemination
- Improvements **also in homogeneous placement**

Thank you.